EAS 487
HOMEWORK 3
SOLUTIONS

1. $\underline{E} = E_0 \left(3\hat{x} + 2e^{j\pi/4}\hat{y} \right) e^{j(\omega t - kz)}$ $\quad \lambda = 1.2\,m \quad E_0 = 1\,mV/m$

a) DIPOLE WILL RECEIVE ALL POWER W $\hat{y}$ LINEAR POLARIZATION

$$E_y = 2E_0 e^{j\pi/4} e^{j(\omega t - kz)}$$

$$P_r = \frac{1}{2}\frac{E_y E_y^*}{Z_0} = \frac{E_0^2}{60\pi} \cong 5\times10^{-9}\ W/m^2 \quad (5.3..)$$

$$A_{EFF} = \frac{\lambda^2}{4\pi D} = \frac{(1.2)^2}{4\pi}1.64 = 0.19\ m^2$$

$$P_{rec} = \frac{E_0^2}{60\pi}\cdot 0.19 \cong \underline{1\times10^{-9}\ W}$$

b) FOR THIS PART, MUST DECOMPOSE WAVE INTO LC & RC COMPONENTS

$$E_0\left(3\hat{x} + 2e^{j\pi/4}\hat{y}\right) = R(\hat{x} - j\hat{y}) + L(\hat{x} + j\hat{y})$$

SO $\quad L + R = 3E_0 \quad (1)$

$$j(L-R) = 2E_0 e^{j\pi/4} = \frac{2E_0}{\sqrt{2}}(1+j)$$

OR $\quad \underline{L - R = \sqrt{2}E_0(1-j)} \quad (2)$

SUM $\quad 2L = E_0\left(3 + \sqrt{2}(1-j)\right)$ OR $\quad L = \frac{E_0}{2}\left(3 + \sqrt{2} - \sqrt{2}j\right)$

$$= E_0\left(2.21 - \frac{\sqrt{2}}{2}j\right)$$ AMPLITUDE OF LC COMPONENT

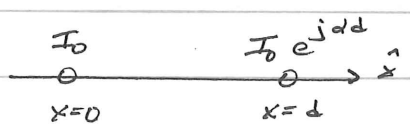
SO LC WAVE LOOKS LIKE

$$E_0\left(2.21 - \frac{\sqrt{2}}{2}j\right)(\hat{x} + j\hat{y})\, e^{j(\omega t - kz)}$$

AND $\quad P_r = \frac{1}{2}\frac{E_L \cdot E_L^*}{Z_0} = \frac{10.8E_0^2}{2Z_0} = 1.43\times10^{-8}\ W/m^2$

$$P_{rec} = P_r \cdot A_{EFF} = 20\cdot 1.43\times10^{-8} = \underline{2.86\times10^{-7}\ W}$$

2



$$|F(\psi)| = I_0 \left| \frac{\sin\left(k\cos\psi + \alpha\right)d}{\sin\left(k\cos\psi + \alpha\right)\frac{d}{2}} \right|$$

$N+1 = 2$

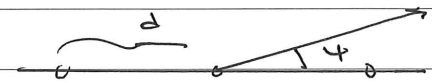
ψ = ANGLE OFF X AXIS

- GET A MAXIMUM AT $\psi = 0$ WITH $\alpha = -k$

$|F|_{MAX} = 2$

- GET A MINIMUM AT $\psi = \pi$ WITH $d = \frac{\lambda}{4}$

2



$$|F(\psi)| = \left| \frac{\sin \frac{N+1}{2}(u+u_0)}{\sin \frac{1}{2}(u+u_0)} \right| \qquad N+1 = 3$$

MAIN BEAM WHERE $kd\cos\psi = u = -u_0 = -\alpha d$

→ LOCATE AT $\psi = 0$ BY MAKING $\alpha = -k = -2\pi/\lambda$

NULLS WHERE $\frac{N+1}{2}(u+u_0) = M\pi$ (BUT NOT IF $\frac{1}{2}(u+u_0) = n\pi$)

OR $kd(\cos\psi - 1) = \frac{2M\pi}{3}$ (BUT NOT IF $kd(\,''\,) = 2n\pi$)

→ LOCATE THESE AT $\psi = \pi$ BY MAKING

$$d = \frac{\lambda}{2\pi}\frac{M\pi}{3} = \frac{M\lambda}{6} \qquad \left(\text{BUT NOT } d = \frac{n\lambda}{2}\right)$$

e.g. $M = 1, 2, 4, 5, \ldots$

ALL TOGETHER, $u + u_0 = kd(\cos\psi - 1) = \frac{2\pi}{\lambda}\frac{M\lambda}{6}(\cos\psi - 1) = \frac{M\pi}{3}(\cos\psi - 1)$

$$|F(\psi)| = \frac{1}{3}\left| \frac{\sin \frac{M\pi}{2}(\cos\psi - 1)}{\sin \frac{M\pi}{6}(\cos\psi - 1)} \right|$$

SEE FIGURE FOR

$M = 1, 2, 4$

→

GRATING LOBES

$M \geq 4$

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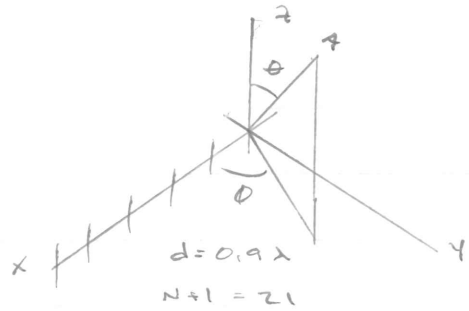
1. 3.



$$|F| = \frac{\left|\sin\left(\frac{N+1}{2}u\right)\right|}{\left|\sin\frac{u}{2}\right|} \qquad u = kd\underbrace{\cos\psi}_{\sin\theta\cos\phi}$$

ARRAY FACTOR

BUT RECALL ELEMENT FACTOR FOR $\frac{\lambda}{2}$ DIPOLE : $\left|\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\right|$ ← OK COORDINATE

$$\text{SO} \quad |F(\theta,\phi)|^2 = \frac{\left|\sin\left(\frac{N+1}{2}kd\sin\theta\cos\phi\right)\right|^2}{\left|\sin\left(\frac{1}{2}kd\sin\theta\cos\phi\right)\right|^2}\left|\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\right|^2$$

* WHICH HAS A MAXIMUM VALUE OF $(N+1)^2$, SO WE SHOULD NORMALIZE BY DIVIDING BY $(N+1)^2$ ACCORDINGLY

PATTERN PLOTS SHOWN FOR Y-Z PLANE (E PLANE) AND X-Y PLANE (H PLANE)

H-PLANE HPBW $2(\pi/2 - 1.54733) = 0.0469$ rad $(2.7°)$

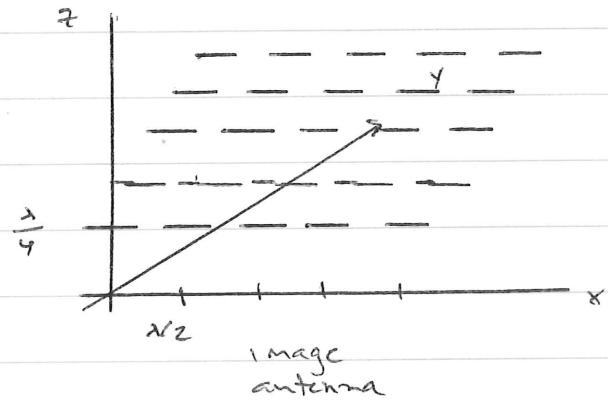
E-PLANE HPBW $2(\pi/2 - 0.88994) = 1.362$ rad $(78°)$

SO COMPUTE $D = \frac{4\pi}{2\cdot HPBW_E \cdot HPBW_H} = \underline{98.36}$ VERY CLOSE TO $5.48\frac{(N+1)d}{\lambda} = \underline{101.6}$

OR DO THIS EXACTLY AS $D = \frac{P_{max}}{P_{total}/4\pi} = \frac{4\pi}{0.162866} = \underline{77.16}$

* FAILURE OF ESTIMATOR IS IN EQUATING SOLID ANGLE Ω WITH PRODUCT OF HALF POWER BEAMWIDTHS. OUR HPBW ESTIMATOR IS QUITE GOOD, HOWEVER.

4



5×5×2 array of $\lambda/2$ dipoles

$u = kd\cos\psi$ ~~[illegible]~~ $u_0 = \alpha d$ — likewise v, v_0

$|F(\psi)| = \frac{1}{n+1}\left|\frac{\sin\frac{n+1}{2}(u+u_0)}{\sin\frac{1}{2}(u+u_0)}\right|$ — w, w_0

i array in x direction: $u = \frac{2\pi}{\lambda}\frac{\lambda}{2}\cos\psi = \pi\sin\theta\cos\phi$; $n+1 = 5$

ii y $v = \quad = \pi\sin\theta\sin\phi$; $n+1 = 5$

iii z $w = \quad = \pi\cos\theta$; $n+1 = 2$

and $\alpha d = \pm\pi$

Dipole element factor $|F(\theta,\phi)| = \frac{\cos(\frac{\pi}{2}\sin\theta\cos\phi)}{\sqrt{1-\sin^2\theta\cos^2\phi}}$ from HW 1

Combine all four terms using pattern multiplication; consider $|F(\theta,\phi)|^2$

a) $P_r \propto |F(\theta,\phi)|^2$ normalize appropriately

b) Numerical calculations give $\theta_{r_2} = 0.175$ rad in E-plane

$= 0.181$ rad in H-plane

corresponding HPBW $= 20°$ E-plane, $20.75°$ H plane

(pretty close)

c) See plots (dB scale)

d) To adjust beam in y-z plane, adjust v_0 s.t.

$v + v_0 = \pi\underbrace{\sin\theta}_{25°}\underbrace{\sin\phi}_{1} + v_0 = 0$

$\Rightarrow v_0 = -0.443\pi$ — See plot (dB scale)

e) θ_{r_2} points now at 0.263, 0.659 rad

So $\text{HPBW}_H = 22.69°$

degraded somewhat by steering